

```
15. # True Range
16. tr = pd.concat([tr1, tr2, tr3], axis=1).max(axis=1)
17.
18. # Average True Range
19. atr = tr.rolling(window=window).mean()
20.
21. return atr
22.
23. # Calculate custom ATR
24. apple['ATR14'] = average_true_range(apple, window=14)
25.
```

Creating custom indicators allows you to:

- **Implement proprietary ideas:** Develop indicators not available in standard libraries
- **Optimize parameters:** Fine-tune indicators for specific markets or conditions
- **Combine indicators:** Create composite indicators that blend multiple signals
- **Understand mechanics:** Gain deeper insight into how indicators actually work

Many successful algorithmic trading strategies rely on proprietary indicators or unique modifications of standard ones, giving them an edge over traders using off-the-shelf solutions.

Identifying Trading Signals

Technical indicators are only valuable if they generate actionable trading signals. Let's explore how to extract these signals from the indicators we've calculated.

Crossover Signals: The Intersection of Lines

Crossovers occur when one line crosses another, potentially signaling a change in trend or momentum. They are among the most commonly used